

Theory Questions

Dataset

1. What is a dataset in data science?

- A **dataset** is a structured collection of data used for analysis, modeling, and decision-making. It can consist of rows and columns (like a table), or more complex formats such as text, images, or videos. In data science, datasets are the foundation for training machine learning models and performing statistical analysis.

2. What are the different types of datasets?

- **Structured dataset** → Organized in rows and columns, easy to store and query (e.g., SQL tables, spreadsheets).
- **Unstructured dataset** → No fixed format, includes text, images, audio, video, social media posts.
- **Semi-structured dataset** → Partially organized with tags or metadata (e.g., JSON, XML, log files).

3. Why is dataset quality important for machine learning?

- High-quality datasets ensure accurate, reliable, and unbiased models. Poor quality data (missing values, duplicates, noise) can lead to incorrect predictions, reduced performance, and misleading insights. Clean, representative datasets improve generalization and trustworthiness of machine learning outcomes.

Data Cleaning

4. What is data cleaning and why is it necessary?

- **Data cleaning** is the process of correcting or removing inaccurate, incomplete, or irrelevant data. It is necessary because raw data often contains errors, duplicates, or inconsistencies that can distort analysis and reduce the accuracy of models.

5. What are common techniques for handling missing values?

- **Deletion** → Remove rows or columns with too many missing values.
- **Imputation** → Fill missing values using mean, median, mode, or predictive models.
- **Interpolation** → Estimate values based on trends or neighboring data points.

6. How can duplicates and inconsistent data types be corrected?

- **Duplicates** → Identify using unique keys or record comparison, then remove or merge.
- **Inconsistent data types** → Standardize formats (e.g., converting strings to dates, integers to floats) to ensure consistency across the dataset.

Exploratory Data Analysis (EDA)

7. What is exploratory data analysis (EDA) and its purpose?

- **Exploratory Data Analysis (EDA)** is the process of summarizing and visualizing datasets to understand their structure, detect patterns, and identify anomalies. Its purpose is to gain insights before applying statistical models or machine learning algorithms.

8. What are univariate, bivariate, and multivariate analyses?

- **Univariate analysis** → Examines one variable at a time (e.g., histogram of age distribution).
- **Bivariate analysis** → Studies relationships between two variables (e.g., scatter plot of height vs. weight).
- **Multivariate analysis** → Explores interactions among three or more variables (e.g., regression analysis with multiple predictors).

9. How does data visualization support EDA?

- **Data visualization** helps present complex data in graphical form, making trends, correlations, and outliers easier to detect. Tools like histograms, scatter plots, box plots, and heatmaps allow analysts to interpret data quickly and communicate findings effectively.

